
Newton School of Technology

Data Visualization & Analytics | Capstone 2 | Concise Final Report

F1 Constructor Sponsorship & Investment ROI Analytics

Easy-to-read decision report for sponsors and evaluators

Team ID Section C — Group G13

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Purpose of this version: This concise report keeps the full business logic, findings, and recommendations of the detailed report, but presents them in a shorter and easier-to-read form.

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1 Executive Overview

This project answers one practical question: **which Formula 1 constructor gives the best return for a sponsor when growth, consistency, risk, and operational quality are considered together?**

Formula 1 is a premium global sponsorship platform, but not every team offers equal value. Some teams deliver strong brand visibility through stable results and growth, while others carry high cost and rising performance risk. This report converts race data into a sponsor decision framework.

Using 2014–2024 Formula 1 data, we built a Python pipeline, cleaned and transformed race-level data, measured five core KPIs, validated key claims with statistics, and delivered three Tableau dashboards.

1.1 Top-Line Conclusions

1. **McLaren is the best current sponsorship opportunity.** It combines the strongest recent growth with strong reliability and the highest Investment Score.
2. **Red Bull remains high-profile but riskier than its recent reputation suggests.** Its trend is negative relative to its previous dominance cycle.
3. **Sauber, Aston Martin, Haas, and Williams offer strong value signals.** These teams appear more attractive for budget-conscious or diversified portfolios.
4. **Operational quality matters.** Reliability and pit-stop execution are not side metrics; they meaningfully explain sponsor-facing performance quality.

Recommended Action

Best strategic approach: build a diversified sponsorship portfolio with one premium growth team, one stable hedge, and one undervalued upside team.

2 Business Problem

Sponsors often choose F1 partnerships based on brand history, visibility, or recent wins. That approach can be misleading because it ignores whether a team is improving, plateauing, or entering decline.

This project solves that problem by replacing intuition with a structured scorecard. The objective is not to predict every race result, but to identify which teams are most likely to deliver consistent brand value.

2.1 Who This Report Is For

This concise report is designed for:

- corporate CMOs and sponsorship decision-makers,
- business evaluators reviewing the project output,
- readers who want the full story without the long technical appendix.

3 Data and Method in Simple Terms

The analysis uses publicly available Formula 1 data from the Ergast historical dataset, focused on the 2014–2024 Hybrid Era. That window was selected because pre-2014 data is not technically comparable to modern Formula 1 in terms of engines, reliability, and competitive structure.

3.1 What Data Was Used

We combined race results, driver data, constructor data, status labels, circuit metadata, and pit-stop records. After filtering and joining, the main race dataset contained 4,626 rows and 35 columns before cleaning and feature engineering.

3.2 What the Pipeline Did

The pipeline followed five stages:

- 1. extract raw CSV files and join the required tables,
- 2. clean missing values and standardise data types,
- 3. engineer features such as finish flag and position delta,
- 4. run EDA and statistical validation,
- 5. export final dashboard-ready files and scorecards.

★ What This Means

The pipeline matters because a sponsorship model is only trustworthy if the underlying race data is clean, consistent, and reproducible.

3.3 ETL Snapshot

Table 1: Simplified ETL summary		
Stage	Output Size	What happened
Raw ingestion	14 CSV files	Historical tables were loaded with null handling for \N values.
Era filter	2014–2024 only	The analysis was restricted to the Hybrid Era for modern comparability.
Master join	4,626 rows	Results, races, constructors, drivers, status, and circuit data were merged.
Cleaning	28 final fields	Sparse columns were dropped and analysis-ready features were created.
Export	4 final CSVs	Tableau-ready files and scorecard outputs were generated for dashboard use.

The main lesson from the ETL stage is that data preparation was not a side task. It shaped the quality of every later result. Without handling null values, choosing the correct modern era, and keeping row-level granularity consistent, the scorecard would not be reliable enough for sponsor decisions.

4 KPI Framework

The report evaluates teams on five dimensions. Together they create a more balanced view than points or wins alone.

Table 2: Five KPI dimensions used in the investment scorecard	
KPI	Why it matters for sponsors
Momentum	Shows whether the team is improving or declining over time.
Consistency	Measures how predictable performance is from race to race.
Reliability	Indicates how often the team finishes and therefore remains visible on race day.
Race Craft	Captures whether teams gain positions during the race itself.
Operational Excellence	Uses pit-stop efficiency and race execution as indicators of team quality.

The final Investment Score combines these measures after normalisation so teams can be compared on one common scale.

4.1 How to Interpret the Scorecard

The scorecard is not meant to replace judgment. Instead, it gives decision-makers a structured first view. If a team scores high on momentum but low on consistency, that may suit a high-risk sponsor but not a brand that needs predictable exposure. If a team scores well on reliability and consistency but less strongly on headline growth, it may still be a very strong hedge within a diversified portfolio.

In other words, the scorecard helps answer both of these questions:

- Which single team looks strongest right now?
- Which mix of teams gives the best balance of upside and stability?

Table 3: Simplified interpretation guide	
Pattern	Sponsor interpretation
High momentum + high consistency	Premium growth opportunity
High momentum + low consistency	High-upside but higher-risk bet
Low momentum + high consistency	Stable but possibly mature sponsorship asset
Low momentum + low consistency	Weak investment case unless priced very cheaply

5 Key Findings

5.1 Finding 1: McLaren Leads the Current Opportunity Set

McLaren achieved the highest Investment Score in the dataset at 63.02 and was the only Gold-tier team. Its year-over-year points growth in 2024 was +128.95%, and its momentum

slope was the strongest positive trend across teams.

Recommended Action

If a sponsor wants one premium team with both credibility and upside, McLaren is the clearest choice.

5.2 Finding 2: Red Bull Is Strong but No Longer the Safest Premium Bet

Red Bull remains a major brand platform, but the model shows deterioration in forward momentum. The team experienced a negative momentum slope and a substantial decline in points relative to its earlier peak.

This does not mean Red Bull is weak. It means the team may now be priced more on legacy dominance than on future efficiency of spend.

5.3 Finding 3: The Grid Is More Open Than Before

The concentration of championship points at the very top has decreased. That means sponsors do not need to pay only for one dominant brand to get strong visibility. Mid-table and upper-midfield teams can now offer better value relative to cost.

5.4 Finding 4: Risk and Stability Matter More Than Reputation

Teams with lower volatility and stronger finish reliability create more dependable sponsor exposure. A team with occasional great results but frequent operational breakdowns may still be a weak sponsorship decision.

5.5 Finding 5: The Model Supports Portfolio Thinking, Not Just Team Ranking

One of the most useful outcomes of the analysis is that it moves the sponsorship conversation away from “Which team is the best?” toward “Which combination of teams gives the best outcome for my budget and risk tolerance?” This is why the second dashboard and the recommendation section matter so much.

For a premium brand, concentration in one elite team may still make sense. For a diversified brand portfolio, however, a combination of one anchor team, one hedge, and one undervalued team is more resilient.

6 Dashboard Reading Guide

This concise report includes the three core dashboards because they are the clearest visual summary of the decision framework.

6.1 Dashboard 1: Constructor Investment Scorecard

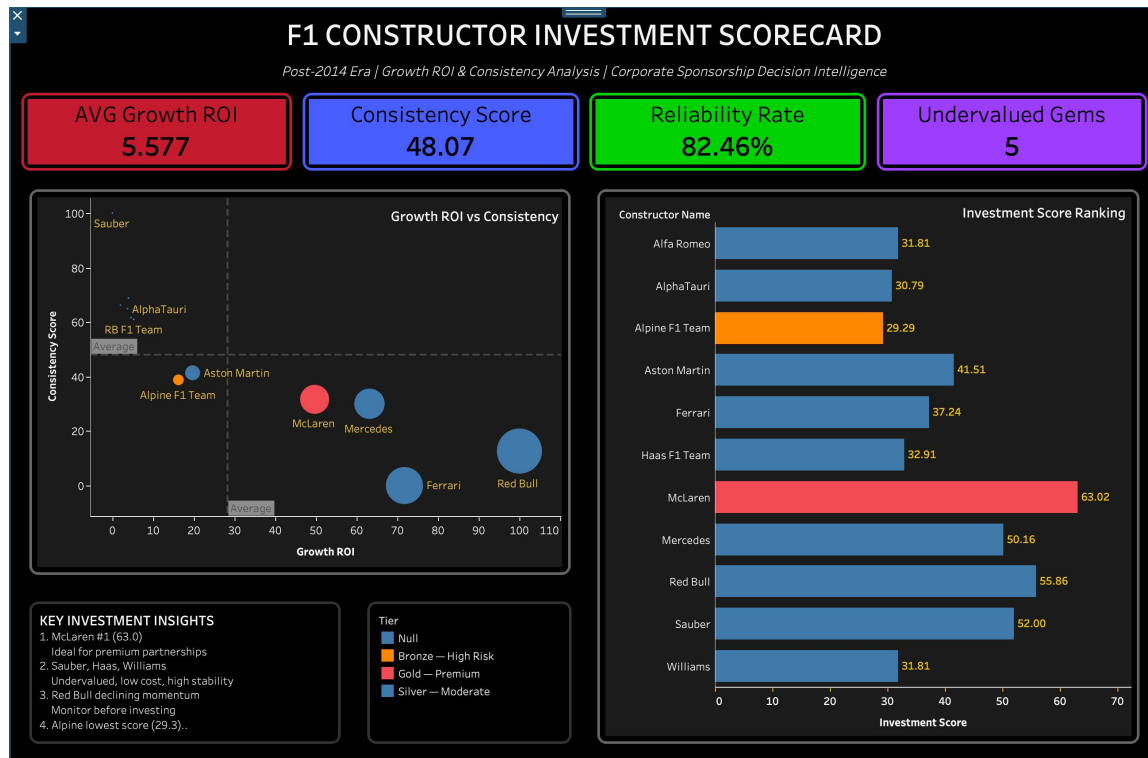


Figure 1: Dashboard 1 — F1 Constructor Investment Scorecard

This dashboard ranks teams by overall Investment Score and places them in a simple growth-versus-consistency view. It is the fastest way to identify premium teams, undervalued teams, and high-risk teams.

6.2 Dashboard 2: Risk-Adjusted Investment Portfolio Matrix

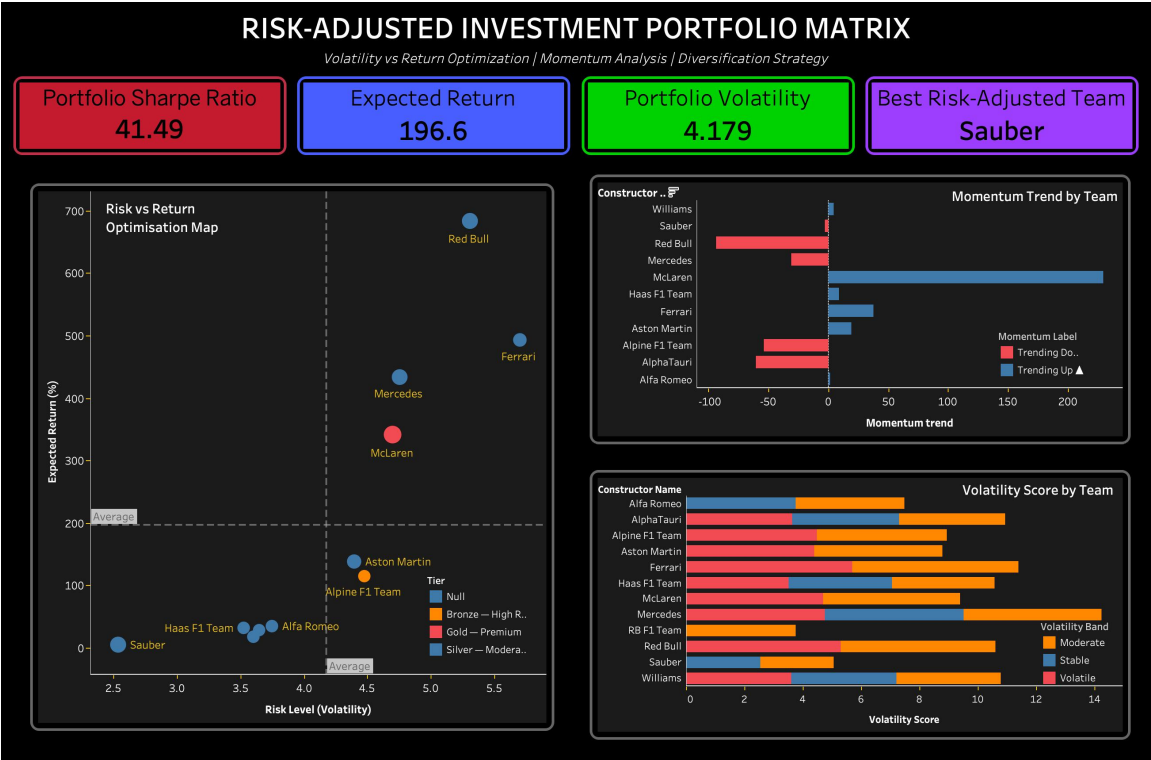


Figure 2: Dashboard 2 — Risk-Adjusted Investment Portfolio Matrix

This dashboard is useful for portfolio thinking. Instead of asking which single team is best, it asks which mix of teams produces the best balance of return and risk.

★ What This Means

The most useful reading of Dashboard 2 is not the single highest-return team. It is the cluster of teams that combine acceptable risk with strong return potential. That is where the report’s diversification logic comes from.

6.3 Dashboard 3: Operational Excellence and Race Craft

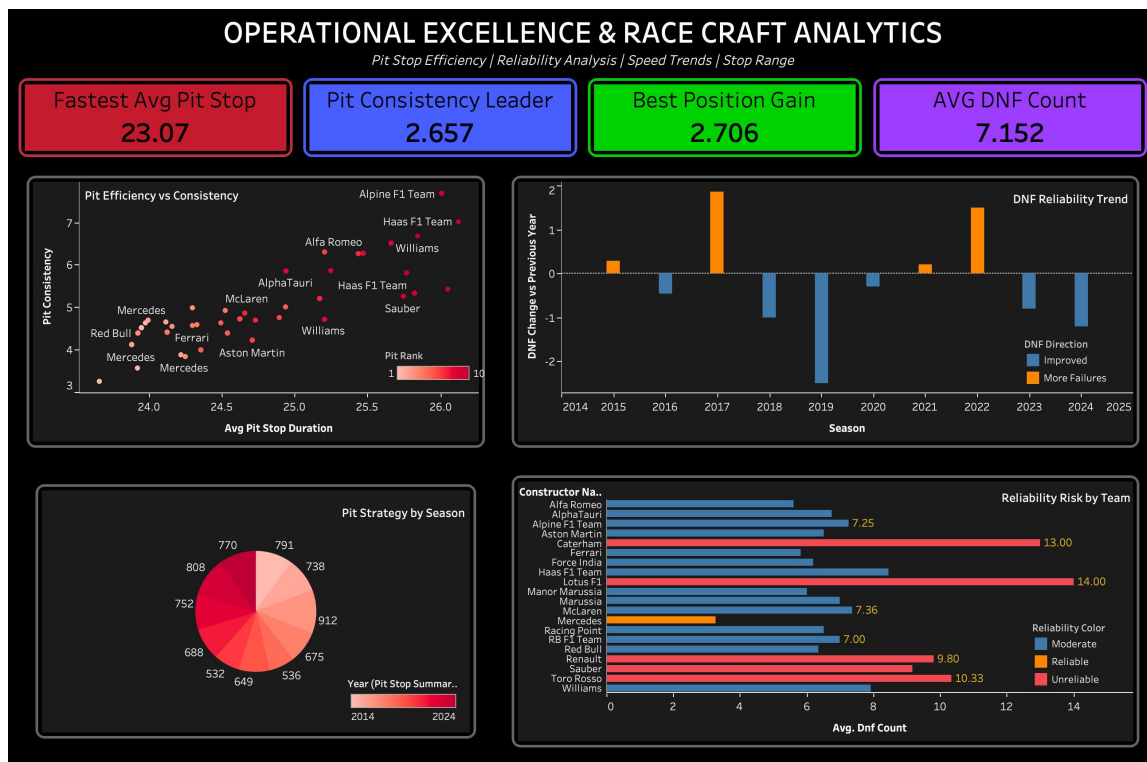


Figure 3: Dashboard 3 — Operational Excellence and Race Craft Analytics

This view focuses on pit-stop efficiency, reliability, and in-race execution. These are especially useful when two teams appear similar on headline results but differ in quality of operations.

6.4 What the Dashboards Add Beyond the Text

The written report explains the logic of the findings, but the dashboards make the model usable. A sponsor or evaluator can quickly see where teams sit on the growth-risk spectrum, how trends are changing, and why operational performance matters. This makes the project not just descriptive, but decision-oriented.

Table 4: What each dashboard contributes

Dashboard	Decision value
Dashboard 1	Ranks teams and identifies premium, moderate, and high-risk opportunities.
Dashboard 2	Helps build diversified sponsor portfolios based on expected return and volatility.
Dashboard 3	Tests whether operational discipline supports the visible race performance of each team.

7 Recommendations

Table 5: Practical sponsor recommendations

Priority	Recommendation	Why it matters
1	Prioritise McLaren for premium partnership discussions.	Best current mix of momentum, reliability, and scorecard performance.
2	Build a diversified F1 portfolio rather than over-concentrating on one team.	Reduces exposure risk while preserving upside.
3	Treat Sauber and Aston Martin as strategic value opportunities.	They offer useful combinations of stability and undervaluation.
4	Review Red Bull deals with caution and performance clauses.	Strong brand, but weaker forward trend than reputation implies.
5	Include operational KPIs in sponsor review frameworks.	Pit-stop speed and reliability reflect team quality and exposure stability.

★ What This Means

The most balanced strategy from this report is: **McLaren as growth anchor, Sauber as stability hedge, and one undervalued team such as Aston Martin or Haas as the upside play.**

7.1 Recommended Sponsor Archetypes

Different sponsors can use the same analysis differently:

Table 6: How different sponsor types can use the model

Sponsor type	Best fit from this report
Premium brand	McLaren as the main visibility and prestige anchor.
Risk-aware portfolio sponsor	McLaren + Sauber + Aston Martin or Haas.
Budget-conscious growth seeker	Target undervalued teams before market repricing.
Performance-sensitive sponsor	Use reliability and pit-stop metrics as contract review triggers.

8 Impact Estimation

Although the dataset does not contain proprietary sponsorship contracts, the model still supports practical business interpretation.

- Premium teams can justify higher rates only if performance trend and operational strength remain high.
- Undervalued teams can offer better return per dollar if their growth is recognised before the market reprices them.

- Reliability matters because every DNF reduces race-day brand visibility.

For a sponsor, this means the decision is not just about fame. It is about buying the right performance stage of a team at the right time.

8.1 What Evaluators Should Notice

From a project-evaluation perspective, the work demonstrates a full analytics cycle:

1. business problem framing,
2. raw data ingestion and cleaning,
3. feature engineering,
4. exploratory and statistical analysis,
5. dashboard creation,
6. final strategic recommendation.

That end-to-end flow is one of the strongest features of the project because it shows both technical and business translation skills.

9 Limitations

This concise report should be read with the following limits in mind:

- It does not use confidential sponsorship fee data.
- It focuses on constructor-level analysis, not individual driver branding.
- It evaluates modern-era F1 only, which is appropriate for current sponsor decisions.
- It treats performance metrics as sponsor-value proxies rather than direct revenue measures.

These limitations do not weaken the usefulness of the report; they simply define the scope of the model.

10 Future Scope

The project can be extended in several practical ways:

- incorporate sponsorship fee estimates or media valuation proxies,
- add qualifying and sprint data for richer weekend analysis,
- build team-level forecasting for the next season,
- add driver-brand fit as a separate marketing dimension,
- convert the model into a reusable sponsor decision tool updated each season.

These extensions would move the project from a decision framework toward a live sponsor intelligence system.

11 Final Conclusion

This project demonstrates that F1 sponsorship decisions can be evaluated more intelligently when teams are measured on growth, consistency, reliability, race execution, and risk

together.

The central conclusion is clear: **McLaren is the strongest current investment-style sponsorship opportunity, while a diversified portfolio approach gives the best balance of exposure and risk control.**

For evaluators, the project shows a complete analytics workflow from raw data to business recommendation. For sponsors, it provides a practical decision tool that is far more useful than choosing only by reputation.

Team Members

Name	Role	Responsibility
Priyanshu Tomar	Team Lead / ETL Lead	Dataset ownership, ETL pipeline, report co-authoring, PPT support
Piyush	Strategy Lead	Recommendation framing, business strategy, PPT co-ownership
Kushagra Maheshwari	EDA & Visualisation Lead	EDA support, dataset support, visual analysis, dashboard coordination
Thejas	Analytics & Stats Lead	Statistical analysis, validation, analytical interpretation
Anurag Kumar	Data Lead / Report Lead	Dataset sourcing, profiling, report co-ownership
Aditya Srivastava	Tableau & PPT Lead	Tableau ownership, presentation design, PPT finalisation

Selected Reference Tables

Table 7: Selected output files generated for the project

File	Rows	Cols	Use
constructor_season_summary.csv	112	21	Team-season scoring and dashboard views
race_detail.csv	4,626	32	Drill-down race analysis
pit_stop_summary.csv	112	9	Operational and pit-stop analysis
investment_scorecard.csv	12	17	Final sponsor ranking and portfolio logic

Table 8: Selected points concentration figures

Year	Top Team	Top Team Share	Top 2 Share
2015	Mercedes	36.63%	58.94%
2021	Mercedes	27.79%	54.31%
2023	Red Bull	34.99%	51.63%
2024	McLaren	25.05%	49.57%

Contribution Snapshot

Table 9: Simplified contribution matrix

Member	Dataset	ETL	EDA	Stats	Tableau	Report	PPT
Priyanshu Tomar	Core	Core	Assist	Assist	Assist	Core	Core
Piyush	Assist	Assist	Assist	Core	Assist	Assist	Core
Kushagra Maheshwari	Assist	Assist	Core	Assist	Core	Assist	Assist
Thejas	Assist	Core	Assist	Core	Assist	Assist	Assist
Anurag Kumar	Core	Assist	Assist	Assist	Assist	Core	Assist
Aditya Srivastava	Assist	Assist	Assist	Assist	Core	Assist	Core

This concise report keeps authorship visible without overwhelming the main business narrative. The full detailed report remains the best source for complete technical ownership and documentation depth.

Quick Glossary

Term	Meaning in this report
Momentum	How strongly a team's points performance is improving or declining over time.
Consistency	How stable the team's race results are from event to event.
Reliability	How often the team finishes races instead of retiring.
Race Craft	How effectively the team gains positions during the race.
Operational Excellence	Team quality reflected in pit stops, execution, and race management.
Investment Score	Combined score built from multiple KPIs to compare sponsor opportunities.
Undervalued Team	A team showing useful performance potential relative to expected market cost.
Risk-Adjusted Portfolio	A sponsor strategy that balances upside with exposure stability.

★ What This Means

If this report is being read quickly by a faculty reviewer or sponsor-minded audience, the glossary helps make the model understandable without needing the full technical document.